See the examples below of how to format the activities, challenges and questions that can feature in workbooks and worksheets

Standard question

What are the two states used in binary? (2 marks)

Scenario-based question

A student has saved revision videos in a folder. The total file size is 4,300 MB. Convert this into GB. (2 marks)

Tables for students to complete

Identify the size of each unit. Complete the table below. (6 marks)

Unit	Number of bits or bytes
Bit	
Nibble	
Byte	
Kilobyte (KB)	
Megabyte (MB)	
Gigabyte (GB)	

Tick-box tables

Tick the correct answer for each question. (5 marks)

Question	Option A	Option B	Option C	Option D
1. Which is the smallest unit of data?	☐ Byte	☐ Bit	☐ KB	☐ MB
2. How many bits are in a nibble?	☐ 2	☐ 4	☐ 8	☐ 16
3. Which conversion is correct?	☐ 3 MB = 300 KB	☐ 3 MB = 3000 KB	☐ 3 MB = 30 KB	☐ 3 MB = 30000 KB
4. Which order is correct from smallest to largest?	☐ GB, MB, KB, Byte	☐ Byte, KB, MB, GB	☐ KB, Byte, MB, GB	☐ MB, GB, Byte, KB
5. Why is binary suitable for computers?	☐ It uses symbols	☐ It has 2 digits	☐ It matches on / off states	☐ It is faster than other systems

Mix and Match

Match each letter below to its correct description. (5 marks)

A	Bit	B	Nibble	C	Byte	D	Megabyte	E	Gigabyte

Letter	**Definition**
	A group of 8 bits
	Used to measure storage on devices such as hard drives
	The smallest unit of data
	A group of 4 bits
	Often used to measure files such as photos and documents

Paragraph Fill-in-the-blanks

Fill in the blank with the correct word from the list below. (6 marks)

List		
divide	multiply	bigger
smaller	same	changes
value	convert	number
stays	unit	size

When converting from one unit to another, the actual amount stays the _________.

Only the metric prefix _________. When converting to a _________ unit, you usually

_________. But when converting to a _________ unit, you usually _________.

Code Fill-in-the-blanks

Fill in the blanks with the correct statement from the list below. (6 marks)

List			
"no"	"yes"	yes	repeat
"bigger"	smaller	"smaller"	converting_to
/	+	*	-
"smaller"	"bigger"	bigger	"yes"
*	+	-	/
value	conversion_factor	answer	repeat

```
value = float(input("Enter the value: "))

conversion_factor = float(input("Enter the conversion factor: "))

repeat = "yes"

while repeat == [1]________:

    converting_to = input("Convert to a smaller or bigger unit? ")

    if converting_to == [2]________:

        answer = value [3]________ conversion_factor

    elif converting_to == [4]________:

        answer = value [5]________ conversion_factor

    print("Converted value:", [6]________)

    repeat = input("Convert again? Enter yes or no: ")
```

Correct the misconception

Read each statement and correct it.

i) “A byte is made up of 4 bits.” (1 mark)

ii) “To convert from KB to MB, you multiply by 1000.” (1 mark)

iii) “Computers use binary because humans prefer the numbers 0 and 1.” (1 mark)

iiii) “When converting units, the amount of data changes.” (1 mark)

Spot the odd one out

Q1) The following options have one option that does not belong.

i) Tick the odd one out and explain why. (2 marks)

- ☐ 8 bits
- ☐ 1 kilobyte
- ☐ 1 byte
- ☐ 2 nibbles

Reason:

____

____

____

ii) Tick the odd one out and explain why. (2 marks)

- ☐ 1500 KB
- ☐ 1.5 MB
- ☐ 0.0015 GB
- ☐ 150 MB

Reason:

____

____

____

iii) Tick the odd one out and explain why. (2 marks)

- ☐ 2.4 GB
- ☐ 2400 MB

- ☐ 0.0024 TB
- ☐ 24 MB

Reason:

Place in to the correct order

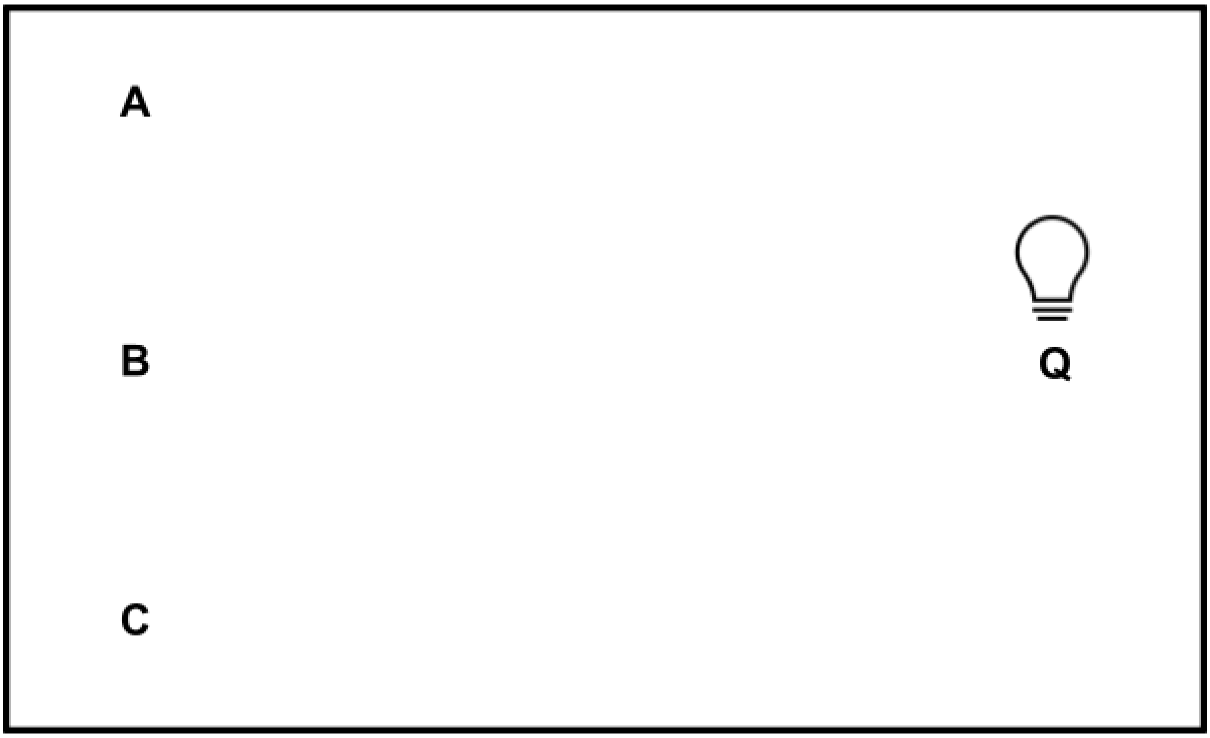
Place in the correct order from biggest to smallest. (5 marks)

A	Bit	B	Nibble	C	Byte	D	Megabyte	E	Gigabyte

Highest
Lowest

Draw a diagram

Q1) Draw the logic gate diagram for Q = A and B or NOT C. (2 Marks)



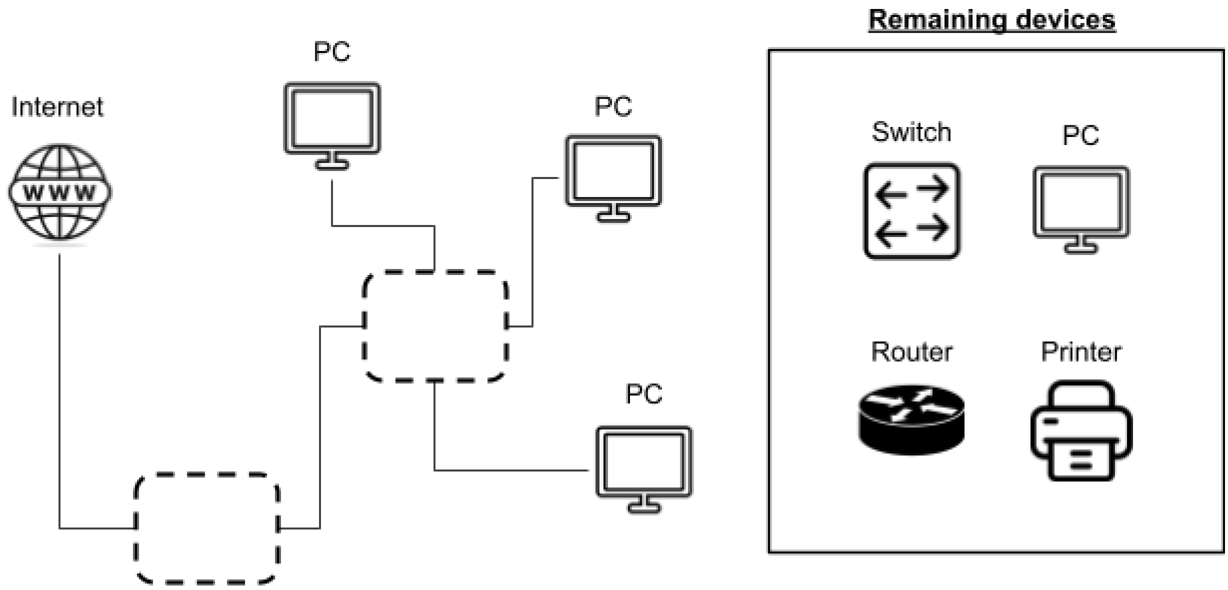
Fill in the truth table

Q1) Complete the truth table for Q = A and B or NOT C. (2 Marks)

A	B	C	A and B	NOT C	Q

Complete the network diagram

Q1) Complete the network diagram by inserting two correct remaining devices.
(2 Marks)



Programming task

Scenario

A teacher is creating a simple after-school club app that will register who is in attendance. To use it teachers will need to login.

Write a program that asks the user to enter a username and password. (6 marks)

Your program should:

1. ask the user to enter a username
2. ask the user to enter a password
3. check whether the username is "admin" and the password is "1234"
4. print "Access granted" if both are correct
5. print "Access denied" if either is incorrect
6. use suitable variable names

Trace table task

Q1) Complete the trace table for the Python program below. (6 marks)

Python
```python
total = 0
for number in range(1, 5):
    total = total + number
    print(total)
```

Line number	total	Output
1	0	-
2		
3		
4		